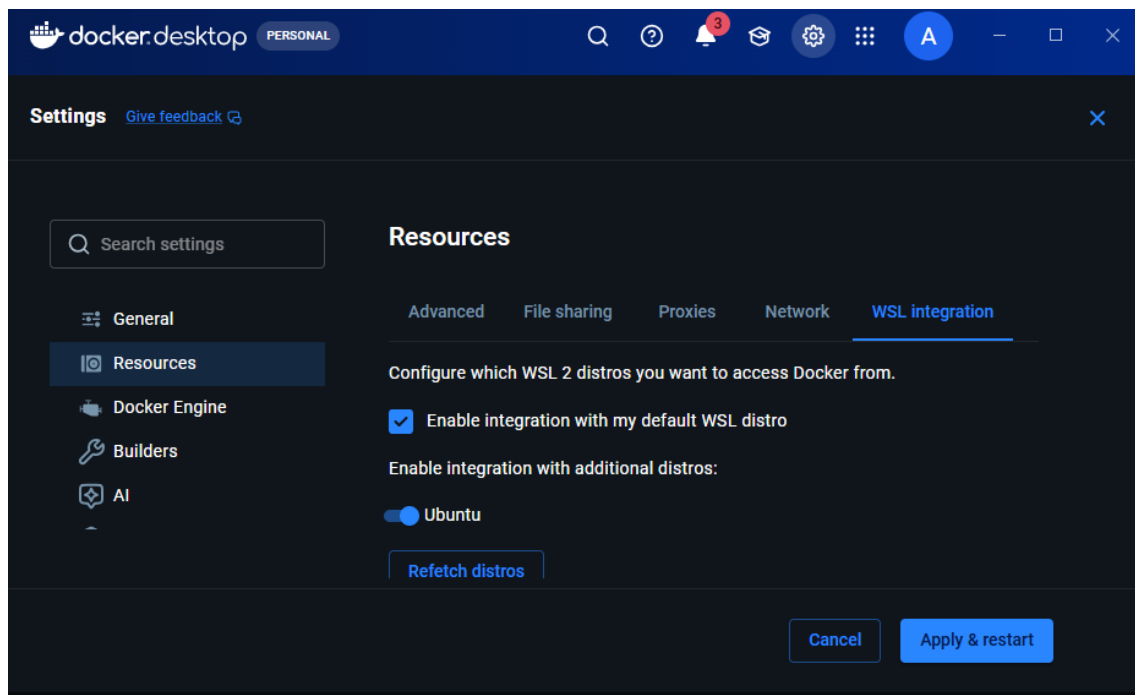


Pulsa en la rueda dentada de docker desktop activa Ubuntu y reinicia la aplicación:



Abrir una terminal con el comando cmd

Entrar en wsl (si no está instalado, instalar con `wsl --install`)

Una vez instalado salir de la distribución Linux con `exit`

Se vuelve a la terminal de Windows

Ejecutar `git clone https://github.com/FlorianFrank/rasta-protocol.git` rasta-protocol-ff

Entrar en distribución Linux con comando `wsl`

Comprobar con el comando `ls` que existe la carpeta `rasta-protocol-ff`

Entrar en la carpeta `/rasta-protocol-ff/src/rasta/c` y ejecutar `vi config.c`

comenta la línea 321 y sustituye el comando por este otro según te muestro:

```
//strncpy(pwrq.ifr_name, ifname, IFNAMSIZ);  
snprintf(pwrq.ifr_name, IFNAMSIZ, "%s", ifname);
```

Pulsar escape :wq! (Para salvar el documento y cerrar el editor)

Volver hasta la carpeta `rasta-protocol-ff` con comando `cd ..`

En dicha carpeta comprobar con comando `ls` que existe el archivo `rasta-docker.sh`

Ejecuta `chmod +x rasta-docker.sh` (modifica el archivo para convertirlo en ejecutable)

Ejecuta ./rasta-docker.sh build y te sale esto por consola:

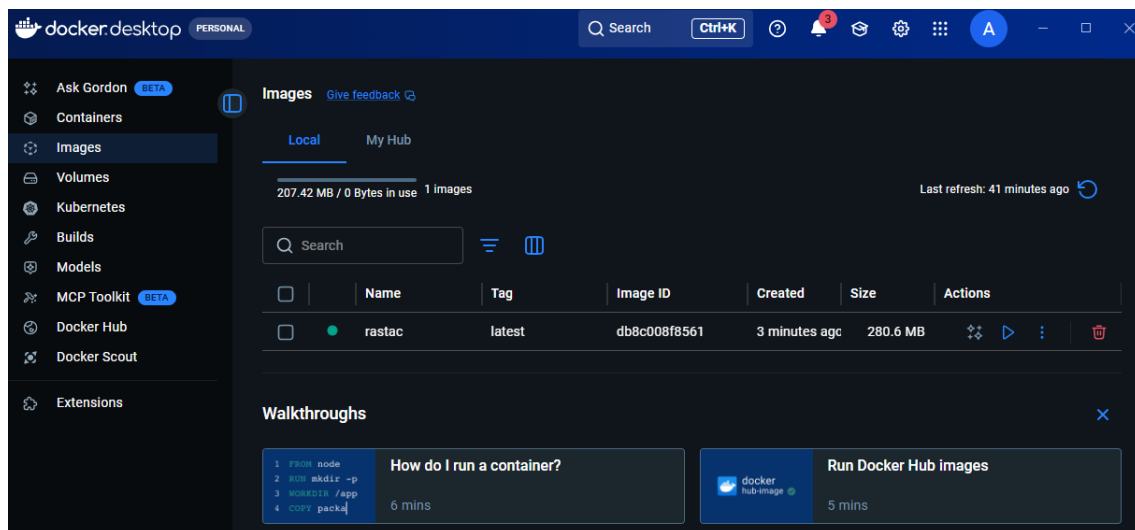
```
Creating network for RaSTA containers...
c946cc132c51e3d34ada81bb9ab8c3b148277de1012d64973105119d80f4b95a
Building container image...
[+] Building 39.4s (18/18) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 1.56kB                               0.1s
=> [internal] load metadata for docker.io/library/alpine:3.12      1.9s
=> [auth] library/alpine:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                   0.1s
=> => transferring context: 2B                                       0.0s
=> [ 1/12] FROM
docker.io/library/alpine:3.12@sha256:c75ac27b49326926b803b9ed43bf088bc220d22556de1b
c5f72d742c913 1.3s
=> => resolve
docker.io/library/alpine:3.12@sha256:c75ac27b49326926b803b9ed43bf088bc220d22556de1b
c5f72d742c91398 0.0s
=> => sha256:1b7ca6aea1ddfe716f3694edb811ab35114db9e93f3ce38d7dab6b4d9270cb0c
2.81MB / 2.81MB          0.9s
=> => extracting
sha256:1b7ca6aea1ddfe716f3694edb811ab35114db9e93f3ce38d7dab6b4d9270cb0c
0.3s
=> [internal] load build context                                  1.0s
=> => transferring context: 506.53kB                                0.9s
=> [ 2/12] RUN apk update && apk add --no-cache cmake gcc make musl-dev linux-headers
14.3s
=> [ 3/12] COPY src/rasta /opt/rasta-c/src/rasta                  0.1s
=> [ 4/12] COPY src/sci /opt/rasta-c/src/sci                      0.1s
=> [ 5/12] COPY src/rasta_example_new /opt/rasta-c/src/rasta_example_new
0.1s
=> [ 6/12] COPY src/scils_example /opt/rasta-c/src/scils_example
0.1s
=> [ 7/12] COPY src/scip_example /opt/rasta-c/src/scip_example
0.1s
=> [ 8/12] COPY src/examples_localhost /opt/rasta-c/src/examples_localhost
0.1s
=> [ 9/12] COPY ./CMakeLists.txt /opt/rasta-c                    0.1s
=> [10/12] COPY config /opt/rasta-c/config                      0.1s
=> [11/12] RUN sed -i -e "s/10.0.0.100/10.10.10.100/g" /opt/rasta-c/config/rasta_server.cfg
&& sed -i -e "s/1 0.3s
=> [12/12] RUN cd /opt/rasta-c/ && mkdir -p build && cd build && cmake -
DEXAMPLE_IP_OVERRIDE:BOOL=ON .. && make 5.3s
=> exporting to image                                           15.1s
=> => exporting layers                                           11.4s
=> => exporting manifest
sha256:e6081b8b800fea0e491a97dfc0553e7ac80fa9ad7de2e3195518b57c2234e528
0.0s
=> => exporting config
sha256:1586f5ca9b8236ff25b3b5e7ca49ab5437abb1e6f6f2dfcade3354ba8f9736a8
0.0s
```

```

=> => exporting attestation manifest
sha256:27d9306f3acca131053d1c1ca710519b6183f5afcdac2ed42cb5c6850eb28116    0.0s
=> => exporting manifest list
sha256:db8c008f8561622f61d03d68b217eb9ef632f3612a884b9d9e4d2f2ab4529806
0.0s
=> => naming to docker.io/library/rastac:latest                                0.0s
=> => unpacking to docker.io/library/rastac:latest                            3.5s
Creating RaSTA containers...
Error response from daemon: No such container: rasta-server
Error response from daemon: No such container: rasta-client1
Error response from daemon: No such container: rasta-client2
Error response from daemon: No such container: rasta-server
Error response from daemon: No such container: rasta-client1
Error response from daemon: No such container: rasta-client2
91a0184a138720856a484db10e7b543d6db6aaac00875fa7d8626af98da5ae55
1a199cb8fda1eccb09885953ab33c8ba3e1dc3b64880147269bc71273afce94b
6ed6bc66a98e1eccb63bdc3a790673937a3a4fa4da4ffb2215fd4ed63e4de9dc

```

Aparece en docker desktop:



Ejecuta `docker exec -it rasta-server sh` (aparecerá este prompt en la consola `/ #`)

Entra en la carpeta examples con este comando `cd opt/rasta-c/build/bin/exe/examples/`

En la pantalla aparece:

```
/opt/rasta-c/build/bin/exe/examples # ./rasta_example r
```

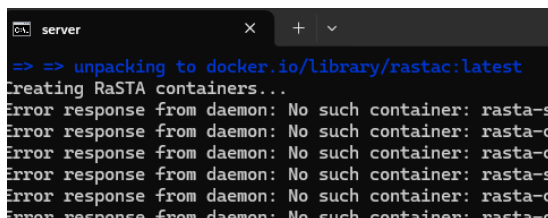
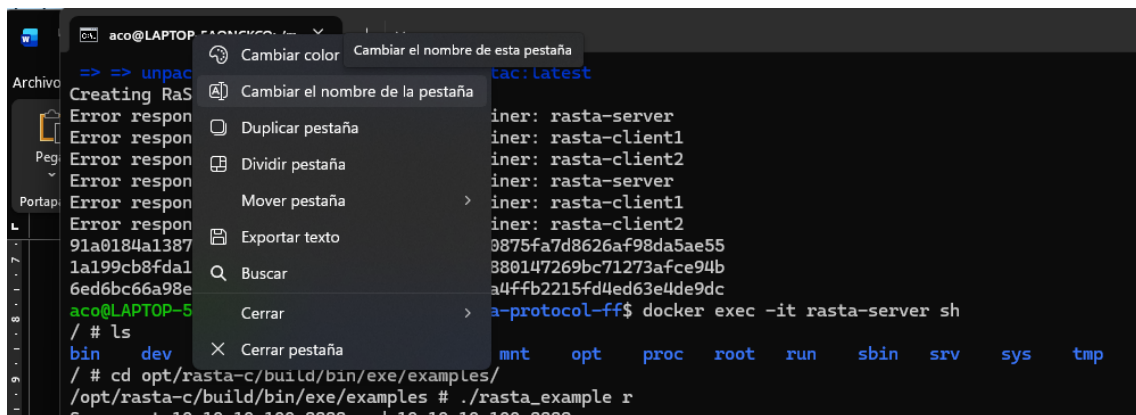
Server at 10.10.10.100:8888 and 10.10.10.100:8889

```
-> R (ID = 0x61)
```

Pulsa enter y sale:

```
-> R (ID = 0x61)
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA HANDLE_INIT] Loaded
accepted version: 0303
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux init] init memory
for 2 listen ports
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux init] loading listen
from config
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux init] initialization
done
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA HANDLE_INIT] Open Mux
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux open] channels : 2
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux open] opening
transport channel 1/2
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux open] listen
thread for port 8888 started
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux open] opening
transport channel 2/2
[03/05/26|09:11:08 (Epoch time: 1772701868956)][DEBUG][RaSTA RedMux receive thread]
Thread 0 running
```

Cambia el nombre a la ventana para que te sea más fácil identificar servidor y cliente pulsando con botón derecho del ratón sobre la pestaña superior de la ventana.



Abre otra terminal de windows:

Ejecuta WSL

entra en el nodo cliente: `docker exec -it rasta-client1 sh` y cambia el nombre a la ventana: client1

```
client1
Microsoft Windows [Versión 10.0.26200.7922]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\Aco>wsl
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

aco@LAPTOP-5AONCKCO:/mnt/c/Users/Aco$ docker exec -it rasta-client1 sh
/ # |
```

Entra en la carpeta examples con este comando `cd opt/rasta-c/build/bin/exe/examples/`

Ejecuta el ejemplo en modo cliente mediante el comando: `./rasta_example s1`

En la terminal aparece:

```
/opt/rasta-c/build/bin/exe/examples # ./rasta_example s1
Server at 10.10.10.100:8888 and 10.10.10.100:8889
-> S1 (ID = 0x62)
-> Press Enter to connect
[03/05/26|09:18:04 (Epoch time: 1772702284333)][DEBUG][RaSTA HANDLE_INIT] Loaded accepted version: 0303
[03/05/26|09:18:04 (Epoch time: 1772702284333)][DEBUG][RaSTA RedMux init] init memory for 2 listen ports
[03/05/26|09:18:04 (Epoch time: 1772702284333)][DEBUG][RaSTA RedMux init] loading listen from config
[03/05/26|09:18:04 (Epoch time: 1772702284333)][DEBUG][RaSTA RedMux init] initialization done
[03/05/26|09:18:04 (Epoch time: 1772702284333)][DEBUG][RaSTA HANDLE_INIT] Open Mux
[03/05/26|09:18:04 (Epoch time: 1772702284333)][DEBUG][RaSTA RedMux open] channels : 2
[03/05/26|09:18:04 (Epoch time: 1772702284333)][DEBUG][RaSTA RedMux open] opening transport channel 1/2
[03/05/26|09:18:04 (Epoch time: 1772702284334)][DEBUG][RaSTA RedMux open] listen thread for port 8890 started
[03/05/26|09:18:04 (Epoch time: 1772702284334)][DEBUG][RaSTA RedMux open] opening transport channel 2/2
[03/05/26|09:18:04 (Epoch time: 1772702284334)][DEBUG][RaSTA RedMux receive thread] Thread 0 running
```

Pulsa una vez enter en la terminal del cliente para que comience con el intercambio de mensajes.

En terminal client1 aparece:

-> Connection request sent to 0x61

Connectionstate change (remote: 97)
CONNECTION_START

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA Red init] space for 2
connected channels

[10/16/25|11:22:28 (Epoch time: 1760613748333)][INFO][RaSTA RedMux add channel] added
new redundancy channel for ID=0x61

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA CONNECT] Using

3406931676 as initial sequence number

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA CONNECT] Local version is 0303

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x61

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA RedMux send] current seq_tx=0

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA RedMux send] redundancy packet created

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.100:8888](#)

[10/16/25|11:22:28 (Epoch time: 1760613748333)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2

Connectionstate change (remote: 97)

CONNECTION_UP

Handshake complete, state is now UP (with ID 0x61)

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RedMux receive] channel 0 received data len = 58

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red receive] Channel 0: ptr=0x5d0aa5fb9020

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red receive] channel 0: correct seq. nr. delivering to next layer

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red receive] channel 0: seq_pdu=0, seq_rx=0

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red receive] channel 0: added message to buffer

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red deliver deferq] f_deliverDeferQueue called

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red deliver deferq] deferq doesn't contain seq_pdu=1

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RedMux receive] channel 1 received data on upd

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RedMux receive thread] Thread 0 calling receive

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux receive] channel 1 received data on upd

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux receive] channel 0 received data len = 44

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux receive] channel 1 received data len = 44

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA Red receive] Channel 0: ptr=0x5d0aa5fb9020

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA Red receive] channel 0: correct seq. nr. delivering to next layer

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA Red receive] channel 0:
seq_pdu=1, seq_rx=1
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA Red receive] channel 0:
added message to buffer
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA Red deliver deferq]
f_deliverDeferQueue called
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA Red deliver deferq] deferq
doesn't contain seq_pdu=2
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux receive] channel 0
received data on upd
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux receive] channel 0
received data len = 44
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA Red receive] Channel 0:
ptr=0x5d0aa5fb9020
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA Red receive] channel 0:
correct seq. nr. delivering to next layer
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA Red receive] channel 0:
seq_pdu=2, seq_rx=2
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux receive] channel 1
received data on upd
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA Red receive] channel 0:
added message to buffer
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux receive] channel 1
received data len = 44
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA Red deliver deferq]
f_deliverDeferQueue called
[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA Red deliver deferq] deferq
doesn't contain seq_pdu=3
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux receive] channel 0
received data on upd
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux receive] channel 0
received data len = 44
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA Red receive] Channel 0:
ptr=0x5d0aa5fb9020
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA Red receive] channel 0:
correct seq. nr. delivering to next layer
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA Red receive] channel 0:
seq_pdu=3, seq_rx=3
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA Red receive] channel 0:
added message to buffer
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA Red deliver deferq]
f_deliverDeferQueue called
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA Red deliver deferq] deferq
doesn't contain seq_pdu=4
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux receive thread]
Thread 0 receive done
[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux receive] channel 1
received data on upd
[10/16/25|11:22:34 (Epoch time: 1760613754417)][DEBUG][RaSTA RedMux receive] Receive
called
[10/16/25|11:22:34 (Epoch time: 1760613754417)][DEBUG][RaSTA RedMux receive] channel 1
waiting for data on fd 4...

[10/16/25|11:22:34 (Epoch time: 1760613754417)][DEBUG][RaSTA RedMux receive] Receive called
[10/16/25|11:22:34 (Epoch time: 1760613754417)][DEBUG][RaSTA RedMux receive] channel 0 waiting for data on fd 3...
[10/16/25|11:22:34 (Epoch time: 1760613754419)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x61
[10/16/25|11:22:34 (Epoch time: 1760613754419)][DEBUG][RaSTA RedMux send] current seq_tx=5
[10/16/25|11:22:34 (Epoch time: 1760613754419)][DEBUG][RaSTA RedMux send] redundancy packet created
[10/16/25|11:22:34 (Epoch time: 1760613754419)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2
[10/16/25|11:22:34 (Epoch time: 1760613754419)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.100:8888](#)
[10/16/25|11:22:34 (Epoch time: 1760613754419)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2

En la terminal del server aparece:

[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd
[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA RedMux receive] channel 0 received data len = 58

Connectionstate change (remote: 98)
CONNECTION_START

[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA RedMux receive] channel 1 received data on upd
[10/16/25|11:22:28 (Epoch time: 1760613748334)][INFO][RaSTA RedMux receive] received pdu from unknown entity with id=0x62
[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA RedMux receive] channel 1 received data len = 58
[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA Red init] space for 2 connected channels
[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA Red receive] Channel 0: ptr=0x580f9d239020
[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA Red receive] channel 0: correct seq. nr. delivering to next layer

Connectionstate change (remote: 98)
CONNECTION_UP

Handshake complete, state is now UP (with ID 0x62)

[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA Red receive] channel 0: seq_pdu=0, seq_rx=0
[10/16/25|11:22:28 (Epoch time: 1760613748334)][DEBUG][RaSTA Red receive] channel 0: added message to buffer
[10/16/25|11:22:28 (Epoch time: 1760613748431)][INFO][RaSTA HANDLE: ConnectionRequest] Add new client 98
[10/16/25|11:22:28 (Epoch time: 1760613748431)][DEBUG][RaSTA HANDLE: ConnectionRequest] Send Connection Response - waiting for Heartbeat

[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)
[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd
[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RedMux receive] channel 0 received data len = 44
[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red receive] Channel 0: ptr=0x580f9d239020
[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA RECEIVE] Received packet 6220 from 98 to 97
[10/16/25|11:22:28 (Epoch time: 1760613748432)][DEBUG][RaSTA Red receive] channel 1: seq_pdu < seq_rx

Received data from Client 98
Packet is from 98
Msg: Message from Sender 1

Send it to other client

[10/16/25|11:22:28 (Epoch time: 1760613748437)][DEBUG][RaSTA RedMux receive] Receive called
[10/16/25|11:22:28 (Epoch time: 1760613748437)][DEBUG][RaSTA RedMux receive] channel 0 waiting for data on fd 3...
[10/16/25|11:22:28 (Epoch time: 1760613748437)][DEBUG][RaSTA RedMux receive] Receive called
[10/16/25|11:22:28 (Epoch time: 1760613748437)][DEBUG][RaSTA RedMux receive] channel 1 waiting for data on fd 4...
[10/16/25|11:22:28 (Epoch time: 1760613748439)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd
[10/16/25|11:22:28 (Epoch time: 1760613748439)][DEBUG][RaSTA RedMux receive] channel 0 received data len = 68
[10/16/25|11:22:28 (Epoch time: 1760613748439)][DEBUG][RaSTA Red receive] Channel 0: ptr=0x580f9d239020
[10/16/25|11:22:28 (Epoch time: 1760613748439)][DEBUG][RaSTA Red receive] channel 0: correct seq. nr. delivering to next layer
[10/16/25|11:22:28 (Epoch time: 1760613748439)][DEBUG][RaSTA Red receive] channel 0: seq_pdu=2, seq_rx=2
[10/16/25|11:22:28 (Epoch time: 1760613748439)][DEBUG][RaSTA Red receive] channel 0: added message to buffer
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x62
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux send] current seq_tx=1
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux send] redundancy packet created
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)
[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8891](#)

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA Red send] Data sent over all transport channels

[10/16/25|11:22:30 (Epoch time: 1760613750410)][DEBUG][RaSTA HEARTBEAT] Heartbeat sent to 98

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA RedMux receive] channel 1 received data on upd

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA RedMux receive] channel 0 received data len = 44

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA RedMux receive] channel 1 received data len = 44

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA Red receive] Channel 0: ptr=0x580f9d239020

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA Red receive] channel 0: correct seq. nr. delivering to next layer

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA Red receive] channel 0: seq_pdu=3, seq_rx=3

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA Red receive] channel 0: added message to buffer

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA Red deliver deferq] f_deliverDeferQueue called

[10/16/25|11:22:30 (Epoch time: 1760613750416)][DEBUG][RaSTA Red deliver deferq] deferq doesn't contain seq_pdu=4

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x62

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux send] current seq_tx=2

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux send] redundancy packet created

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8891](#)

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA Red send] Data sent over all transport channels

[10/16/25|11:22:32 (Epoch time: 1760613752411)][DEBUG][RaSTA HEARTBEAT] Heartbeat sent to 98

[10/16/25|11:22:32 (Epoch time: 1760613752417)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd

[10/16/25|11:22:32 (Epoch time: 1760613752428)][DEBUG][RaSTA RECEIVE] Waiting for incoming packets...

[10/16/25|11:22:32 (Epoch time: 1760613752428)][DEBUG][RaSTA RedMux retrieve all] waiting for message on any connection

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x62

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux send] current seq_tx=3

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux send] redundancy packet created

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8891](#)

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA Red send] Data sent over all transport channels

[10/16/25|11:22:34 (Epoch time: 1760613754412)][DEBUG][RaSTA HEARTBEAT] Heartbeat sent to 98

[10/16/25|11:22:34 (Epoch time: 1760613754419)][DEBUG][RaSTA RedMux receive] channel 0 received data on upd

[10/16/25|11:22:34 (Epoch time: 1760613754424)][DEBUG][RaSTA RedMux receive] Receive called

[10/16/25|11:22:34 (Epoch time: 1760613754424)][DEBUG][RaSTA RedMux receive] Receive called

[10/16/25|11:22:34 (Epoch time: 1760613754424)][DEBUG][RaSTA RedMux receive] channel 1 waiting for data on fd 4...

[10/16/25|11:22:34 (Epoch time: 1760613754424)][DEBUG][RaSTA RedMux receive] channel 0 waiting for data on fd 3...

[10/16/25|11:22:34 (Epoch time: 1760613754429)][DEBUG][RaSTA RECEIVE] Waiting for incoming packets...

[10/16/25|11:22:34 (Epoch time: 1760613754429)][DEBUG][RaSTA RedMux retrieve all] waiting for message on any connection

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x62

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA RedMux send] current seq_tx=4

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA RedMux send] redundancy packet created

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8891](#)

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA Red send] Data sent over all transport channels

[10/16/25|11:22:36 (Epoch time: 1760613756413)][DEBUG][RaSTA HEARTBEAT] Heartbeat sent to 98

[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x62

[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA RedMux send] current seq_tx=5

[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA RedMux send] redundancy packet created
[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2
[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)
[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2
[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8891](#)
[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA Red send] Data sent over all transport channels
[10/16/25|11:22:38 (Epoch time: 1760613758414)][DEBUG][RaSTA HEARTBEAT] Heartbeat sent to 98
[10/16/25|11:22:40 (Epoch time: 1760613760415)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x62
[10/16/25|11:22:40 (Epoch time: 1760613760415)][DEBUG][RaSTA RedMux send] current seq_tx=6
[10/16/25|11:22:40 (Epoch time: 1760613760415)][DEBUG][RaSTA RedMux send] redundancy packet created
[10/16/25|11:22:40 (Epoch time: 1760613760415)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2
[10/16/25|11:22:40 (Epoch time: 1760613760416)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)
[10/16/25|11:22:40 (Epoch time: 1760613760416)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2
[10/16/25|11:22:40 (Epoch time: 1760613760416)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8891](#)
[10/16/25|11:22:40 (Epoch time: 1760613760416)][DEBUG][RaSTA Red send] Data sent over all transport channels
[10/16/25|11:22:40 (Epoch time: 1760613760416)][DEBUG][RaSTA HEARTBEAT] Heartbeat sent to 98
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA RedMux send] sending a data packet to id 0x62
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA RedMux send] current seq_tx=7
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA RedMux send] redundancy packet created
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA RedMux send] Sending on transport channel 1/2
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8890](#)
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA RedMux send] Sending on transport channel 2/2
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA RedMux send] Sent data over channel [10.10.10.110:8891](#)
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA Red send] Data sent over all transport channels
[10/16/25|11:22:42 (Epoch time: 1760613762418)][DEBUG][RaSTA HEARTBEAT] Heartbeat sent to 98

Cuando dejas de ejecutar el cliente aparece en la terminal del server:

Entity 0x62 had a heartbeat timeout!

Connectionstate change (remote: 98)
CONNECTION_CLOSED

```
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA HEARTBEAT] T_i timer
expired - send DisconnectionRequest
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA RedMux send] sending a
data packet to id 0x62
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA RedMux send] current
seq_tx=8
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA RedMux send] redundancy
packet created
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA RedMux send] Sending on
transport channel 1/2
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA RedMux send] Sent data
over channel 10.10.10.110:8890
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA RedMux send] Sending on
transport channel 2/2
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA RedMux send] Sent data
over channel 10.10.10.110:8891
[10/16/25|11:22:44 (Epoch time: 1760613764419)][DEBUG][RaSTA Red send] Data sent over
all transport channels
[10/16/25|11:22:44 (Epoch time: 1760613764420)][DEBUG][RaSTA RedMux remove channel]
removing channel with ID=0x62
```

Otras pruebas que puedes hacer:

Antes de ejecutar los ejemplos de servidor y cliente puedes comprobar la dirección ip de cada nodo con el comando ifconfig

```
/opt/rasta-c/build/bin/exe/examples # ifconfig
eth0      Link encap:Ethernet  HWaddr 72:63:06:6B:95:DB
          inet addr:10.10.10.100  Bcast:10.10.10.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:57 errors:0 dropped:0 overruns:0 frame:0
          TX packets:33 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:4870 (4.7 KiB)  TX bytes:2878 (2.8 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

También puedes comprobar la comunicación ip con el comando ping .

Por ejemplo, desde la consola del servidor puedes hacer ping al cliente 1 así:

```
/opt/rasta-c/build/bin/exe/examples # ping 10.10.10.110
PING 10.10.10.110 (10.10.10.110): 56 data bytes
64 bytes from 10.10.10.110: seq=0 ttl=64 time=0.208 ms
64 bytes from 10.10.10.110: seq=1 ttl=64 time=0.093 ms
64 bytes from 10.10.10.110: seq=2 ttl=64 time=0.071 ms
64 bytes from 10.10.10.110: seq=3 ttl=64 time=0.088 ms
64 bytes from 10.10.10.110: seq=4 ttl=64 time=0.104 ms
64 bytes from 10.10.10.110: seq=5 ttl=64 time=0.070 ms
64 bytes from 10.10.10.110: seq=6 ttl=64 time=0.070 ms
64 bytes from 10.10.10.110: seq=7 ttl=64 time=0.071 ms
64 bytes from 10.10.10.110: seq=8 ttl=64 time=0.069 ms
64 bytes from 10.10.10.110: seq=9 ttl=64 time=0.079 ms
^X64 bytes from 10.10.10.110: seq=10 ttl=64 time=0.065 ms
^C
```

Pruébalo y me dices si tienes algún problema.

Saludos!